

Provable LLM Tamper Detection: Catching Model Provider Tampering of Open-Weights LLMs

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Abstract

1 INTRODUCTION

2 RELATED WORK

3 PROBLEM SETTING

3.1 THREAT MODEL

Throughout this paper, we consider the scenario of a user who accesses a given open-weights LLM through a provider’s black-box inference API. Examples of this service model include NVIDIA-hosted NIM endpoints, Fireworks AI Serverless, and Together AI Serverless Endpoints [NVIDIA, 2026, Fireworks AI, 2026, Together AI, 2026]. The user’s objective is to determine whether the model configuration they are being served has been tampered with. Examples of tampering include serving a different model than the one advertised, injecting an undisclosed prompt, serving an undeclared fine-tune, or using an undeclared quantization scheme.

In this threat model, we define an *untampered* model configuration as one that satisfies the following desirable properties:

1. Generation uses a standard randomized sampling decoder: generation is nondeterministic, and repeated identical requests are independent samples from a fixed conditional output distribution.
2. The user can reconstruct the full token sequence of the LLM’s context window immediately before the first token is generated.
3. The user can reconstruct the full sequence of the LLM’s generated tokens.
4. The served architecture, weights, and tokenizer are identical to those of the specified public model checkpoint, except for weight quantization permitted by property 5.
5. The weight quantization method satisfies an expected standard (e.g. unquantized, or a quantization scheme reported by the provider).
6. The provider only uses standard inference-time decoding parameters, which are defined in Section 3.3.

7. The underlying model configuration being served, as well as the decoding parameter values, remain fixed across all queries.

A model configuration that does not satisfy the definition of *untampered* is considered *tampered*.

We furthermore make the following assumptions:

1. The user has full access to the provider model’s advertised open weights checkpoint.
2. The user can submit arbitrarily many queries to the black-box provider’s service.
3. The user can observe the string representation of any returned response from the provider, though not necessarily the individual token boundaries or IDs.
4. The user can restrict each returned response from the provider to a chosen maximum number of output tokens.

3.2 THREAT MODEL DISCUSSION

3.2.1 RATIONALE

In the definition of *untampered*, properties 1–5 and 7 define the baseline for everyday faithful LLM serving: ordinary stochastic inference, input/output visibility, fidelity to the advertised model, acceptable quantization, and consistency across queries. Furthermore, because the full universe of possible decoding parameters is unstandardized and potentially infinite, property 6 requires that the provider’s decoding parameters be drawn from a canonical set of recognized, widely-used decoding parameters. This prevents an adversarial provider who uses obscure or proprietary decoding parameters to maliciously manipulate the model’s outputs from satisfying the definition of *untampered* by technicality. To restrict the scope of tampering to more invasive forms of model configuration manipulation (e.g. fine-tuning, stealthy quantization), the specific choices of values among the standard decoding parameters will not be considered tampering.

3.2.2 IMPLICATIONS FOR MALICIOUS PROVIDERS

In this setting, a malicious provider’s primary option for plausibly denying tampering is to explain away unexpected measured model behavior as being caused by benign hidden values of standard decoding parameters. In 2026, the growing prevalence of model APIs with unexposed decoding parameters in releases from major LLM vendors makes this defense increasingly credible. For example, OpenAI does not expose temperature or top- p for GPT-5.2 in thinking mode; Anthropic does not expose temperature, top- p , or top- k to the user on Claude 4.7 and later; Google does not expose temperature, top- p , or top- k for Gemini 3 models; and DeepSeek does not expose temperature, top- p , presence penalty, or frequency penalty in thinking mode [OpenAI, n.d.b, Anthropic, n.d., Google Cloud, 2026, DeepSeek, n.d.]. The central question of this paper is whether a user can demonstrate a model configuration to be tampered despite the obfuscation and plausible deniability afforded by hidden standard decoding parameter values.

3.3 STANDARD DECODING PARAMETERS

Here, we compile a list of standard decoding parameters found in widely-used LLM APIs and inference stacks, including OpenAI’s API, Hugging Face Transformers, llama.cpp, and KoboldCpp [OpenAI, n.d.a, Hugging Face, n.d.b, ggml-org, 2026b, LostRuins, 2026]. We refer to the parameters on this list as the *standard decoding parameters* throughout the paper.

Let z_t be the model’s raw logit for token t at the first output position, and let $p_t = \exp(z_t) / \sum_s \exp(z_s)$ be its probability. The parameters below transform these logits or probabilities before the first output token is sampled. Excluding token t means setting $z_t = -\infty$ and renormalizing the remaining probabilities.

- **Temperature.** For $T > 0$, replace z_t by z_t/T .
- **Logit bias or token bias.** For a fixed per-token bias b_t , replace z_t by $z_t + b_t$ [OpenAI, n.d.a].
- **Top- p (nucleus sampling).** Sort tokens by decreasing probability and retain the shortest prefix whose cumulative mass is at least p [Holtzman et al., 2020].
- **Top- k .** Retain the k tokens with greatest probability [Hugging Face, n.d.b].
- **Frequency penalty.** If c_t is the number of occurrences of t in the history consulted by the decoder, replace z_t by $z_t - \lambda_f c_t$.
- **Presence penalty.** Replace z_t by $z_t - \lambda_p \mathbf{1}\{c_t > 0\}$.
- **Repetition penalty.** For a penalty $r > 0$, rescale logits of tokens found in the consulted history using the implementation’s fixed sign-aware rule; the standard Transformers rule divides a positive logit by r and multiplies a negative logit by r [Hugging Face, 2026].
- **Structured-output enforcement.** Given a grammar or schema, retain exactly the tokens that can validly extend it from the current prefix [ggml-org, 2026a,b].
- **No-repeat- n -gram.** Exclude every token that would complete an n -gram already present in the generated history [Hugging Face, n.d.b].
- **Min- p .** With $p_{\max} = \max_t p_t$, retain tokens satisfying $p_t \geq \alpha p_{\max}$ [Nguyen et al., 2025].
- **Typical- p .** Sort tokens by increasing $|\log p_t - H(p)|$ and retain the shortest prefix whose cumulative probability is at least the typical- p threshold [Meister et al., 2023].
- **Mirostat.** Maintain a feedback state targeting a specified mean surprise and use it to adapt the top- k cutoff [Basu et al., 2021].
- **Mirostat 2.** Maintain an adaptive threshold μ targeting mean surprise τ . Exclude tokens with $-\log p_t > \mu$, sample from the remaining tokens, and update μ using the difference between the sampled token’s surprise and τ [Basu et al., 2021].
- **η -sampling.** For hyperparameter ϵ , let $\eta = \min\{\epsilon, \sqrt{\epsilon} \exp(-H(p))\}$ and retain tokens with $p_t > \eta$ [Hewitt et al., 2022].
- **Tail-free sampling.** Sort tokens by decreasing probability and compute the normalized magnitudes of the second finite differences of those probabilities. Retain tokens until their cumulative value reaches the threshold z , and exclude all remaining tokens [Bricken, 2019, LostRuins, 2026].
- **Top- a .** With $p_{\max} = \max_t p_t$, retain tokens satisfying $p_t \geq a p_{\max}^2$ [LostRuins, 2026].

4 CONVENTIONS

Nondegeneracy. Throughout this paper, our results tacitly assume natural nondegeneracy and regularity conditions when appropriate. We state these conditions explicitly whenever they are nontrivial.

Token identification. To account for tokenizer differences across model configurations, we identify tokens by their string representations rather than by their underlying token IDs, unless stated otherwise.

5 DEFINITIONS

Expected and served models. Let M denote the *expected model*: the open-weights architecture, checkpoint weights at their published precision, and tokenizer that the user believes the provider is serving. Let $\widetilde{M}$ denote the *served model*: the architecture, weights, and tokenizer actually used by the provider.

Decoding parameters. For any token t , let $b(t)$ denote the token-specific logit bias used by the provider’s API, and let $T > 0$ denote the temperature used by the provider’s API.

Multinomial experiments. A *multinomial experiment* (or simply an *experiment* when the meaning is clear) consists of repeatedly submitting the same request to the provider’s LLM API and recording the string representation of the first generated token returned from each query. The user can observe this token without knowing its underlying token ID by limiting each response to one generated token.

Each multinomial experiment E_j has the following associated quantities:

- An expected context C_j , chosen by the experimenter, representing the context that the user *believes* the served LLM $\widetilde{M}$ receives immediately before the first token is generated.
- Given any token t , $z_j(t)$, the raw logit assigned to t by the expected model M on context C_j , and $\widetilde{z}_j(t)$, the corresponding raw logit assigned by the served model $\widetilde{M}$ on the context actually used by the provider. The quantity $z_j(t)$ is computable by the user, whereas $\widetilde{z}_j(t)$ is unobserved.
- A number of queries N_j , chosen by the experimenter.
- Two token lists $A_j = (a_{1,j}, \dots, a_{k_j,j})$ and $B_j = (b_{1,j}, \dots, b_{k_j,j})$, where both k_j and the tokens in each list are chosen by the experimenter.
- For each $i = 1, \dots, k_j$, the observed counts $n_j(a_{i,j})$ and $n_j(b_{i,j})$, giving the number of queries whose first generated token returned was $a_{i,j}$ or $b_{i,j}$, respectively.
- For each $i = 1, \dots, k_j$, the unobserved theoretical quantities $p_j(a_{i,j})$ and $p_j(b_{i,j})$, giving the true probabilities that the first generated token returned by a query is $a_{i,j}$ or $b_{i,j}$, respectively. ($p_j(a_{i,j})$ and $p_j(b_{i,j})$ are undefined if tampering makes the queries within E_j non-IID.)

Remark. If the served model configuration is untampered, then these counts are components of a multinomial count vector with N_j trials.

Example. Consider a hypothetical model that uses the ChatML template [Hugging Face, n.d.a]. With an empty system message and the assistant generation prompt included, let C_j be the following string:

```
<|im_start|>system
<|im_end|>
<|im_start|>user
Generate a random number from the set {zero, one}. Output a digit and
nothing else.<|im_end|>
<|im_start|>assistant
```

Take $N_j = 4$, $k_j = 1$, $A_j = (a_{1,j}) = ("0")$, and $B_j = (b_{1,j}) = ("1")$. If the first tokens of the four same-query results are "0", "1", "1", "one", then $n_j(a_{1,j}) = 1$ and $n_j(b_{1,j}) = 2$; the final string belongs to neither list.

Derived quantities. For each multinomial experiment E_j , define the logit aggregates $z_{a,j} = \sum_{i=1}^{k_j} z_j(a_{i,j})$, $z_{b,j} = \sum_{i=1}^{k_j} z_j(b_{i,j})$, and $\Delta z_j = z_{a,j} - z_{b,j}$; the bias aggregates $b_{a,j} = \sum_{i=1}^{k_j} b(a_{i,j})$, $b_{b,j} = \sum_{i=1}^{k_j} b(b_{i,j})$, and $\Delta b_j = b_{a,j} - b_{b,j}$; and the log-ratios $\gamma_j = \ln\left(\prod_{i=1}^{k_j} \frac{n_j(a_{i,j})}{n_j(b_{i,j})}\right)$ and $\Gamma_j = \ln\left(\prod_{i=1}^{k_j} \frac{p_j(a_{i,j})}{p_j(b_{i,j})}\right)$ (Γ_j is undefined if tampering makes the queries within E_j non-IID).

ρ -regularity. An experiment E_j is ρ -regular if the tokens in A_j and B_j are distinct and, for every $t \in A_j \cup B_j$: (1) t belongs to M 's vocabulary; (2) t does not occur in C_j under M 's tokenizer; and (3) $p_j(t) > 0$, meaning that t is able to be returned by the provider's LLM API during experiment E_j .

Remark. In practice, fulfilling condition 2 only requires the experimenter to choose C_j , A_j , and B_j such that no token in $A_j \cup B_j$ is a substring of C_j . Furthermore, condition 3 can be fulfilled by querying the provider's LLM API a small number of times and choosing the tokens for A_j and B_j from among the first generated tokens returned.

6 GENERAL FORM OF THE DETECTION METHOD

We first present the general forms of the estimator test and its equivalent hypothesis test for detecting provider-side model configuration tampering. We will later derive concrete special cases of this form and then demonstrate its efficacy in proof-of-concept experiments.

Smooth aggregate estimators. Let $\phi : \mathbb{R}^m \rightarrow \mathbb{R}$ be smooth. For ρ -regular experiments $E_1, \dots, E_m$, define $\hat{\phi} := \phi(\gamma_1, \dots, \gamma_m)$ and $\phi^* := \phi(\Gamma_1, \dots, \Gamma_m)$ (ϕ^* is undefined if tampering makes the queries within any E_j non-IID).

Theorem 1 (General asymptotic properties). *If the provider's model configuration is untampered and the experiments are conducted independently, then, as every $N_j \rightarrow \infty$: $\hat{\phi}$ is a consistent estimator of ϕ^* ; $\hat{\phi}$ is asymptotically normal; $\text{Var}(\hat{\phi}) = \Theta(\sum_{j=1}^m N_j^{-1})$; and $\text{Bias}(\hat{\phi}) = O(\sum_{j=1}^m N_j^{-1})$.*

The proof is deferred to Appendix A.1.

Remark. If the N_j are fixed positive proportions of the total query count $N = \sum_{j=1}^m N_j$, then $\text{Var}(\hat{\phi}) = \Theta(N^{-1})$. Thus, the estimator achieves the optimal $1/N$ variance-decay rate available to regular estimators [Hájek, 1970].

Remark. The standard deviation $\text{SD}(\hat{\phi}) = \Theta(\sqrt{\sum_{j=1}^m N_j^{-1}})$ asymptotically dominates the bias $\text{Bias}(\hat{\phi}) = O(\sum_{j=1}^m N_j^{-1})$. Thus, we may treat the bias as negligible in downstream asymptotic analyses.

Tamper-detection tests. We call $(\hat{\phi}, C)$, where $C \in \mathbb{R}$ and $\hat{\phi}$ is as defined above, a *tamper-detection test* if, whenever the provider's model configuration is untampered, $\hat{\phi}$ consistently estimates C as every $N_j \rightarrow \infty$.

Remark. By Theorem 1, if the provider's model configuration is untampered, then $(\hat{\phi}, \phi^*)$ is a tamper-detection test. However, because whether the model configuration is tampered is unknown, ϕ^* cannot

generally serve as a known reference value. Thus, this construction cannot be used in general. In later sections, we show how to construct tamper-detection tests with explicitly known reference values.

Detection rule (estimator version). Suppose $(\hat{\phi}, C)$ is a tamper-detection test. We say that a model configuration is *detected via estimation* if $\hat{\phi}$ fails to converge to C as every $N_j \rightarrow \infty$.

Theorem 2 (100% long-run detection precision guarantee: estimator version). *If a model configuration is detected via estimation, then it is always tampered.*

Proof. If the model configuration were untampered, $\hat{\phi}$ would converge to C by the definition of a tamper-detection test and hence would not be detected via estimation. The claim follows by contraposition. $\square$

Detection rule (hypothesis-test version). Let $(\hat{\phi}, C)$ be a tamper-detection test. Choose any function $\epsilon(N_1, \dots, N_m) > 0$ such that $\epsilon \rightarrow 0$ and $\epsilon / \sqrt{\sum_{j=1}^m N_j^{-1}} \rightarrow \infty$ as every $N_j \rightarrow \infty$. We say that a model configuration is *detected via hypothesis testing at* $(N_1, \dots, N_m)$ if $|\hat{\phi} - C| > \epsilon(N_1, \dots, N_m)$ for fixed $N_1, \dots, N_m$.

Example. One valid choice is $\epsilon(N_1, \dots, N_m) = \min(N_1, \dots, N_m)^{-\alpha}$ for any $0 < \alpha < 1/2$.

Theorem 3 (100% long-run detection precision guarantee: hypothesis-test version). *As every $N_j \rightarrow \infty$, the probability of an untampered model configuration being falsely detected via hypothesis testing converges to 0.*

The proof is deferred to Appendix A.2.

7 THEORETICAL PROPERTIES OF UNTAMPERED MODEL CONFIGURATIONS

Throughout this section, the served model configuration is assumed to be *untampered*. We further fix two ρ -regular experiments E_1 and E_2 such that $A_1 = A_2 = (a_1, \dots, a_k)$ and $B_1 = B_2 = (b_1, \dots, b_k)$. Because the two experiments share token lists, $\Delta b_1 = \Delta b_2$; we denote their common value by Δb .

7.1 PAIRWISE-RATIO INVARIANCE

Theorem 4 (Pairwise-ratio invariance). *For each $j \in \{1, 2\}$ and $i \in \{1, \dots, k\}$, the ratio $p_j(a_i)/p_j(b_i)$ is invariant under every standard decoding parameter except temperature T and the token biases $b(a_i)$ and $b(b_i)$.*

Proof. Fix $j \in \{1, 2\}$ and $i \in \{1, \dots, k\}$. First consider top- p , top- k , structured-output enforcement, no-repeat- n -gram, min- p , typical- p , Mirostat, Mirostat 2, η -sampling, tail-free sampling, and top- a . Each of these parameters retains a subset S of tokens and renormalizes their probabilities without changing the relative probabilities of the retained tokens. Because E_j is ρ -regular, both a_i and b_i survive every such restriction, and the common renormalization cancels from their probability ratio. Thus, these eleven parameters do not affect it.

It remains to consider frequency penalty, presence penalty, and repetition penalty. By ρ -regularity, neither a_i nor b_i occurs in the context, so these parameters leave both logits unchanged. Since these logits are used in the calculation of the token probabilities, $p_j(a_i)/p_j(b_i)$ is also unchanged.

The only remaining standard decoding parameters are temperature and token bias, which proves the claim. $\square$

Corollary 5. For each $j \in \{1, 2\}$, $\prod_{i=1}^k \frac{p_j(a_i)}{p_j(b_i)}$, and therefore Γ_j , is invariant under every standard decoding parameter except temperature and token bias. □

Proof. Apply Theorem 4 to each factor in the product. □

Corollary 6. For each $j \in \{1, 2\}$, $\Gamma_j = (\Delta z_j + \Delta b)/T$.

Proof. For each i , Theorem 4 allows $p_j(a_i)/p_j(b_i)$ to be computed using only temperature and token bias. For any token t , $p_j(t) = \exp((z_j(t) + b(t))/T) / \sum_s \exp((z_j(s) + b(s))/T)$, so $p_j(a_i)/p_j(b_i) = \exp((z_j(a_i) - z_j(b_i) + b(a_i) - b(b_i))/T)$. Taking the product over i and then its logarithm proves the claim. □

7.2 TEMPERATURE ESTIMATION

Theorem 7. The provider’s temperature satisfies $T = (\Delta z_1 - \Delta z_2)/(\Gamma_1 - \Gamma_2)$.

Proof. Subtracting Corollary 6 for $j = 2$ from the same identity for $j = 1$ gives $\Gamma_1 - \Gamma_2 = (\Delta z_1 - \Delta z_2)/T$. □

Theorem 8. As $N_1, N_2 \rightarrow \infty$, $(\Delta z_1 - \Delta z_2)/(\gamma_1 - \gamma_2)$ is a consistent estimator of T .

Proof. This is the special case of Theorem 1 with $\phi(x_1, x_2) = (\Delta z_1 - \Delta z_2)/(x_1 - x_2)$. □

7.3 LOGIT BIAS DELTA ESTIMATION

Theorem 9. The aggregate token-bias difference satisfies $\Delta b = (\Gamma_2 \Delta z_1 - \Gamma_1 \Delta z_2)/(\Gamma_1 - \Gamma_2)$.

Proof. Substituting Corollary 6 into the right-hand side and simplifying gives Δb . □

Theorem 10. As $N_1, N_2 \rightarrow \infty$, $(\gamma_2 \Delta z_1 - \gamma_1 \Delta z_2)/(\gamma_1 - \gamma_2)$ is a consistent estimator of Δb .

Proof. This is the special case of Theorem 1 with $\phi(x_1, x_2) = (x_2 \Delta z_1 - x_1 \Delta z_2)/(x_1 - x_2)$. □

Remark. Taking $k = 1$ and repeating the construction for multiple token pairs recovers their relative token biases. In particular, fixing a reference token r allows $b(t) - b(r)$ to be estimated for each target token t . Therefore, token biases are identifiable up to a common additive constant; if an anchor such as a known zero-bias token is available, their absolute values are also recoverable.

8 CONCRETE TAMPER-DETECTION TEST EXAMPLES

In this section, we construct two simple concrete examples of the tamper-detection tests defined in Section 6 (General Form of the Detection Method). The key idea for both is this: if the model configuration is untampered, the results of the previous section guarantee that reconstructing the same quantity in two separate ways yields the same value. Thus, inconsistency implies tampering.

Two-experiment estimators. If E_{j_1} and E_{j_2} are ρ -regular experiments satisfying $A_{j_1} = A_{j_2}$ and $B_{j_1} = B_{j_2}$, then we define two estimators:

$$\begin{aligned}\widehat{T}_{j_1, j_2} &:= \frac{\Delta z_{j_1} - \Delta z_{j_2}}{\gamma_{j_1} - \gamma_{j_2}}, \\ \widehat{\Delta b}_{j_1, j_2} &:= \frac{\gamma_{j_2} \Delta z_{j_1} - \gamma_{j_1} \Delta z_{j_2}}{\gamma_{j_1} - \gamma_{j_2}}.\end{aligned}$$

We immediately obtain the following two tamper-detection tests, which inherit the 100% long-run detection precision guarantees from Theorems 2 and 3.

8.1 TEMPERATURE TEST

Theorem 11. *Let E_i, E_j, E_k, E_ℓ be ρ -regular experiments satisfying $A_i = A_j$, $B_i = B_j$, $A_k = A_\ell$, and $B_k = B_\ell$. Then $(\widehat{T}_{i,j} - \widehat{T}_{k,\ell}, 0)$ is a valid tamper-detection test.*

Proof. Applying Theorem 8 to (E_i, E_j) and (E_k, E_ℓ) , if the provider's model configuration is untampered, then $\widehat{T}_{i,j}$ and $\widehat{T}_{k,\ell}$ both consistently estimate T . Their difference therefore consistently estimates 0, so the result follows from the definition of a tamper-detection test. $\square$

Remark. Except where doing so makes an estimator degenerate, the indices i, j, k, ℓ need not be distinct.

8.2 LOGIT BIAS DELTA TEST

Theorem 12. *Let E_i, E_j, E_k, E_ℓ be ρ -regular experiments satisfying $A_i = A_j = A_k = A_\ell$ and $B_i = B_j = B_k = B_\ell$. Then $(\widehat{\Delta b}_{i,j} - \widehat{\Delta b}_{k,\ell}, 0)$ is a valid tamper-detection test.*

Proof. Applying Theorem 10 to (E_i, E_j) and (E_k, E_ℓ) , if the provider's model configuration is untampered, then $\widehat{\Delta b}_{i,j}$ and $\widehat{\Delta b}_{k,\ell}$ both consistently estimate Δb . Their difference therefore consistently estimates 0, so the result follows from the definition of a tamper-detection test. $\square$

Remark. Except where doing so makes an estimator degenerate, the indices i, j, k, ℓ need not be distinct.

8.3 ON RECALL

Although perfect long-run detection recall is not guaranteed for these tests, it is intuitive why a malicious provider may find them difficult to spoof: (1) the many possible tests force an effective spoof to satisfy a complex constraint satisfaction problem; and (2) making the two separate temperature (or logit-bias-delta) estimates equal would likely reflect either an implausible coincidence or deliberate engineering. We furthermore present a toy argument in Appendix A.3 that is suggestive of the difficulty of strong adversarial spoofing, but we do not make claims about its universality.

9 EMPIRICAL PROOF OF CONCEPT

10 CONCLUSION

For expository clarity, we associated each experiment E_j with a single γ_j that used all elements of A_j and B_j . This restriction was stronger and more practically limiting than was logically required by our proofs. A single

experiment E_j can instead produce multiple γ statistics, each using only subsets of A_j and B_j , provided that the one-to-one correspondence between the selected a 's and b 's is preserved. Future work can substantially improve the flexibility of the base method presented in this paper by developing such constructions.

AI USE STATEMENT

GPT-5.6 Sol was used to edit the exposition of this paper and generate boilerplate code for the proof-of-concept experiments. The research problem, core ideas, theoretical results, proofs, and overall research direction were developed by the author. All AI-assisted text and code were reviewed and verified by the author, who takes full responsibility for the final content of this work.

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A APPENDIX

A.1 GENERAL ASYMPTOTIC PROPERTIES

Proof of consistency. Fix $r \in \{1, \dots, m\}$. For each $\ell \in \{1, \dots, k_r\}$, the multinomial law gives $n_r(a_{\ell,r})/N_r \rightarrow p_r(a_{\ell,r})$ and $n_r(b_{\ell,r})/N_r \rightarrow p_r(b_{\ell,r})$. Hence $n_r(a_{\ell,r})/n_r(b_{\ell,r}) = [n_r(a_{\ell,r})/N_r]/[n_r(b_{\ell,r})/N_r] \rightarrow p_r(a_{\ell,r})/p_r(b_{\ell,r})$. Taking the product over ℓ and then its logarithm gives $\gamma_r \rightarrow \Gamma_r$. Since r was arbitrary and ϕ is smooth and therefore continuous, $\hat{\phi} = \phi(\gamma_1, \dots, \gamma_m) \rightarrow \phi(\Gamma_1, \dots, \Gamma_m) = \phi^*$, proving consistency.

Proof of asymptotic normality. First, note that by the law of large numbers, for every r and i , $n_r(a_{i,r})/N_r \rightarrow p_r(a_{i,r})$ and $n_r(b_{i,r})/N_r \rightarrow p_r(b_{i,r})$ as $N_r \rightarrow \infty$. Define $X_{i,r} = n_r(a_{i,r})/N_r$ and $Y_{i,r} = n_r(b_{i,r})/N_r$.

Thus far, we have been treating the γ_r as the atomic building blocks of $\phi(\gamma_1, \dots, \gamma_m)$, but there are objects that are even more granular: the $n_r(a_{i,r})$ and $n_r(b_{i,r})$. Recall that

$$\gamma_r = \ln \left(\prod_{i=1}^{k_r} \frac{n_r(a_{i,r})}{n_r(b_{i,r})} \right) = \sum_{i=1}^{k_r} \ln n_r(a_{i,r}) - \sum_{i=1}^{k_r} \ln n_r(b_{i,r}).$$

Thus, we may view $\phi(\gamma_1, \dots, \gamma_m)$ as a function of the $n_r(a_{i,r})$ and $n_r(b_{i,r})$, or equivalently as a function of the $X_{i,r}$ and $Y_{i,r}$.

We wish to apply the delta method to $\phi(\gamma_1, \dots, \gamma_m)$ as a function of the $X_{i,r}$ and $Y_{i,r}$, but we must first ensure that it is a fixed function that does not depend on the N_r . Fix r and consider index i in the summation above. Then

$$\ln n_r(a_{i,r}) - \ln n_r(b_{i,r}) = \ln(N_r X_{i,r}) - \ln(N_r Y_{i,r}) = \ln X_{i,r} - \ln Y_{i,r}.$$

Thus, each term in the summation defining γ_r does not depend on N_r . Therefore, each γ_r , and hence $\phi(\gamma_1, \dots, \gamma_m)$, is a fixed function of the $X_{i,r}$ and $Y_{i,r}$.

We may now apply the delta method:

$$\begin{aligned} \phi(\gamma_1, \dots, \gamma_m) &\approx c + \sum_{r=1}^m \sum_{i=1}^{k_r} d_{i,r} (X_{i,r} - p_r(a_{i,r})) \\ &\quad + \sum_{r=1}^m \sum_{i=1}^{k_r} e_{i,r} (Y_{i,r} - p_r(b_{i,r})) \end{aligned}$$

as $N_1, \dots, N_m \rightarrow \infty$, for some fixed constants c , $d_{i,r}$, and $e_{i,r}$. For each r , the quantity

$$\sum_{i=1}^{k_r} d_{i,r} X_{i,r} + \sum_{i=1}^{k_r} e_{i,r} Y_{i,r}$$

is independent of the corresponding quantities for the other experiments. Therefore, to prove asymptotic normality of $\phi(\gamma_1, \dots, \gamma_m)$, it suffices to prove that this quantity is asymptotically normal for each r .

We have

$$\sum_{i=1}^{k_r} d_{i,r} X_{i,r} + \sum_{i=1}^{k_r} e_{i,r} Y_{i,r} = \sum_{i=1}^{k_r} d_{i,r} \frac{n_r(a_{i,r})}{N_r} + \sum_{i=1}^{k_r} e_{i,r} \frac{n_r(b_{i,r})}{N_r}.$$

Write each $n_r(a_{i,r})$ as a sum of IID indicator random variables $I_{q,r}(a_{i,r})$ for $q = 1, \dots, N_r$, which record whether $a_{i,r}$ was the first generated token returned on query q of experiment E_r . Define $I_{q,r}(b_{i,r})$ similarly. We now have

$$\begin{aligned} &\sum_{i=1}^{k_r} d_{i,r} \frac{n_r(a_{i,r})}{N_r} + \sum_{i=1}^{k_r} e_{i,r} \frac{n_r(b_{i,r})}{N_r} \\ &= \frac{1}{N_r} \sum_{q=1}^{N_r} \left(\sum_{i=1}^{k_r} d_{i,r} I_{q,r}(a_{i,r}) + \sum_{i=1}^{k_r} e_{i,r} I_{q,r}(b_{i,r}) \right). \end{aligned}$$

Take

$$Z_{q,r} = \sum_{i=1}^{k_r} d_{i,r} I_{q,r}(a_{i,r}) + \sum_{i=1}^{k_r} e_{i,r} I_{q,r}(b_{i,r}).$$

Note that the $Z_{q,r}$ are IID. The expression then becomes $N_r^{-1} \sum_{q=1}^{N_r} Z_{q,r}$, which is asymptotically normal by the central limit theorem. This completes the proof.

Proof of the variance rate. From the proof of asymptotic normality, we see that as $N_1, \dots, N_m \rightarrow \infty$, $\phi(\gamma_1, \dots, \gamma_m)$ can be written, up to an additive constant, as

$$\sum_{r=1}^m \left(\sum_{i=1}^{k_r} d_{i,r} X_{i,r} + \sum_{i=1}^{k_r} e_{i,r} Y_{i,r} \right) = \sum_{r=1}^m \left(\frac{1}{N_r} \sum_{q=1}^{N_r} Z_{q,r} \right).$$

Since the experiments are statistically independent, we have

$$\begin{aligned} \text{Var}(\phi(\gamma_1, \dots, \gamma_m)) &= \text{Var} \left(\sum_{r=1}^m \frac{1}{N_r} \sum_{q=1}^{N_r} Z_{q,r} \right) \\ &= \sum_{r=1}^m \text{Var} \left(\frac{1}{N_r} \sum_{q=1}^{N_r} Z_{q,r} \right). \end{aligned}$$

Since the $Z_{q,r}$ are IID for any given r , give $Z_{q,r}$, $q = 1, \dots, N_r$, the common variance v_r . Then

$$\sum_{r=1}^m \text{Var} \left(\frac{1}{N_r} \sum_{q=1}^{N_r} Z_{q,r} \right) = \sum_{r=1}^m \frac{v_r}{N_r} = \Theta \left(\sum_{r=1}^m \frac{1}{N_r} \right).$$

Proof of the bias rate. First, we show that $\mathbb{E}[\gamma_r] - \Gamma_r = O(N_r^{-1})$ for every r . Recall that

$$\gamma_r = \sum_{i=1}^{k_r} \ln X_{i,r} - \sum_{i=1}^{k_r} \ln Y_{i,r}.$$

Then

$$\begin{aligned} \mathbb{E}[\gamma_r] - \Gamma_r &= \sum_{i=1}^{k_r} \mathbb{E}[\ln X_{i,r}] - \sum_{i=1}^{k_r} \mathbb{E}[\ln Y_{i,r}] \\ &\quad - \left(\sum_{i=1}^{k_r} \ln p_r(a_{i,r}) - \sum_{i=1}^{k_r} \ln p_r(b_{i,r}) \right) \\ &= \sum_{i=1}^{k_r} (\mathbb{E}[\ln X_{i,r}] - \ln p_r(a_{i,r})) - \sum_{i=1}^{k_r} (\mathbb{E}[\ln Y_{i,r}] - \ln p_r(b_{i,r})). \end{aligned}$$

Since k_r is fixed, it suffices to show that each $\mathbb{E}[\ln X_{i,r}] - \ln p_r(a_{i,r})$ and each $\mathbb{E}[\ln Y_{i,r}] - \ln p_r(b_{i,r})$ term is $O(N_r^{-1})$. We show this for $\mathbb{E}[\ln X_{i,r}] - \ln p_r(a_{i,r})$; the proof for the $Y_{i,r}$ case is identical.

We have by the 2nd-order Taylor approximation that:

$$\ln X_{i,r} = \ln p_r(a_{i,r}) + \frac{X_{i,r} - p_r(a_{i,r})}{p_r(a_{i,r})} - \frac{(X_{i,r} - p_r(a_{i,r}))^2}{2p_r(a_{i,r})^2} + R_{i,r}.$$

Thus, we have:

$$\begin{aligned} \mathbb{E}[\ln X_{i,r}] - \ln p_r(a_{i,r}) &= \frac{1}{p_r(a_{i,r})} \mathbb{E}[X_{i,r} - p_r(a_{i,r})] \\ &\quad - \frac{1}{2p_r(a_{i,r})^2} \mathbb{E}[(X_{i,r} - p_r(a_{i,r}))^2] + \mathbb{E}[R_{i,r}]. \end{aligned}$$

Here are some standard facts from binomial distributions:

- $\mathbb{E}[(X_{i,r} - p_r(a_{i,r}))^2] = O(N_r^{-1})$.
- $\mathbb{E}[|X_{i,r} - p_r(a_{i,r})|^3] = O(N_r^{-3/2})$.

Note that the first term is simply 0. Furthermore, the second term is $O(N_r^{-1})$. $|R_{i,r}| = O(|X_{i,r} - p_r(a_{i,r})|^3)$ follows from a standard bound for Taylor polynomial remainders. Thus,

$$|\mathbb{E}[R_{i,r}]| \leq \mathbb{E}[|R_{i,r}|] = O(\mathbb{E}[|X_{i,r} - p_r(a_{i,r})|^3]).$$

Therefore, $|\mathbb{E}[R_{i,r}]| = O(N_r^{-3/2}) = O(N_r^{-1})$. Therefore, $\mathbb{E}[\ln X_{i,r}] - \ln p_r(a_{i,r}) = O(N_r^{-1})$. This proves that $\mathbb{E}[\gamma_r] - \Gamma_r = O(N_r^{-1})$.

Let $\gamma = (\gamma_1, \dots, \gamma_m)^\top$ and $\Gamma = (\Gamma_1, \dots, \Gamma_m)^\top$. By the 2nd-order Taylor approximation, we have that:

$$\phi(\gamma) = \phi(\Gamma) + \nabla \phi(\Gamma)^\top (\gamma - \Gamma) + \frac{1}{2}(\gamma - \Gamma)^\top H_\phi(\Gamma)(\gamma - \Gamma) + R.$$

Thus, we have:

$$\begin{aligned} \mathbb{E}[\phi(\gamma)] - \phi(\Gamma) &= \nabla \phi(\Gamma)^\top \mathbb{E}[\gamma - \Gamma] \\ &\quad + \mathbb{E}\left[\frac{1}{2}(\gamma - \Gamma)^\top H_\phi(\Gamma)(\gamma - \Gamma)\right] + \mathbb{E}[R]. \end{aligned}$$

We have

$$\nabla \phi(\Gamma)^\top \mathbb{E}[\gamma - \Gamma] = O\left(\sum_{r=1}^m \frac{1}{N_r}\right)$$

because $\mathbb{E}[\gamma_r] - \Gamma_r = O(N_r^{-1})$ for every r . We have:

$$\begin{aligned} \left| \mathbb{E}\left[\frac{1}{2}(\gamma - \Gamma)^\top H_\phi(\Gamma)(\gamma - \Gamma)\right] \right| &\leq \mathbb{E}\left[\frac{1}{2} \left| (\gamma - \Gamma)^\top H_\phi(\Gamma)(\gamma - \Gamma) \right| \right] \\ &\leq \frac{1}{2} \|H_\phi(\Gamma)\|_2 \mathbb{E}[\|\gamma - \Gamma\|_2^2] \\ &= \text{const} \cdot \sum_{r=1}^m \mathbb{E}[(\gamma_r - \Gamma_r)^2]. \end{aligned}$$

We then have

$$\begin{aligned} \mathbb{E}[(\gamma_r - \Gamma_r)^2] &= \text{Var}(\gamma_r - \Gamma_r) + \mathbb{E}[\gamma_r - \Gamma_r]^2 \\ &= \text{Var}(\gamma_r) + O(N_r^{-2}) \quad (\text{since } \mathbb{E}[\gamma_r] - \Gamma_r = O(N_r^{-1})) \\ &= O(N_r^{-1}) + O(N_r^{-2}) \\ &= O(N_r^{-1}). \end{aligned}$$

Here, $\text{Var}(\gamma_r) = O(N_r^{-1})$ follows by applying the preceding variance calculation to the identity estimator for E_r . Thus,

$$\left| \mathbb{E}\left[\frac{1}{2}(\gamma - \Gamma)^\top H_\phi(\Gamma)(\gamma - \Gamma)\right] \right| = O\left(\sum_{r=1}^m \frac{1}{N_r}\right).$$

$|R| = O(\|\gamma - \Gamma\|_2^3)$ follows from a standard bound for Taylor polynomial remainders. Then

$$\begin{aligned} |\mathbb{E}[R]| &\leq \mathbb{E}[|R|] \\ &= \mathbb{E}[O(\|\gamma - \Gamma\|_2^3)] \\ &\leq \mathbb{E}\left[O\left(\left(\sum_{r=1}^m |\gamma_r - \Gamma_r|\right)^3\right)\right]. \end{aligned}$$

The multinomial third-moment bound, applied to the smooth log-ratio defining γ_r , gives $\mathbb{E}[|\gamma_r - \Gamma_r|^3] = O(N_r^{-3/2})$. Since $(\sum_{r=1}^m x_r)^3 \leq m^2 \sum_{r=1}^m x_r^3$ for nonnegative x_r ,

$$\begin{aligned} |\mathbb{E}[R]| &= O\left(\sum_{r=1}^m \mathbb{E}[|\gamma_r - \Gamma_r|^3]\right) \\ &= O\left(\sum_{r=1}^m N_r^{-3/2}\right) \\ &= O\left(\sum_{r=1}^m N_r^{-1}\right), \end{aligned}$$

where m is fixed. Thus, $|\mathbb{E}[R]| = O(\sum_{r=1}^m N_r^{-1})$. We conclude that

$$\mathbb{E}[\phi(\gamma)] - \phi(\Gamma) = O\left(\sum_{r=1}^m \frac{1}{N_r}\right).$$

This completes the proof.

A.2 LONG-RUN DETECTION PRECISION FOR HYPOTHESIS TESTING

Proof of Theorem 3. Assume that the provider's model configuration is untampered. By the definition of a tamper-detection test, $\hat{\phi} \rightarrow C$ as every $N_j \rightarrow \infty$. Theorem 1 also gives $\hat{\phi} \rightarrow \phi^*$, so $C = \phi^*$. The theorem further gives that $\hat{\phi}$ is asymptotically normal and $\text{SD}(\hat{\phi}) = \Theta(\sqrt{\sum_{j=1}^m N_j^{-1}})$. Furthermore, its standard deviation asymptotically dominates its bias, so the bias is negligible here.

Let $X \sim \mathcal{N}(0, \sigma^2)$. Fix $p \in (1/2, 1)$, and let $q_p(\sigma) = \inf\{x : \Pr(X \leq x) \geq p\}$ be the p -quantile of X . Then $q_p(\sigma) = z_p \sigma$, where $z_p = \Phi^{-1}(p)$ and Φ is the standard normal CDF. Indeed, for $Z \sim \mathcal{N}(0, 1)$, $\Pr(X \leq x) = \Pr(Z \leq x/\sigma)$, so $\Pr(X \leq q_p(\sigma)) = p$ is equivalent to $\Pr(Z \leq q_p(\sigma)/\sigma) = p$, and hence $q_p(\sigma)/\sigma = \Phi^{-1}(p)$.

Apply this calculation to the asymptotic normal approximation of $\hat{\phi} - \mathbb{E}[\hat{\phi}]$, with $\sigma = \text{SD}(\hat{\phi})$. Its $(1-p)$ - and p -quantiles are therefore asymptotic to $z_{1-p} \text{SD}(\hat{\phi})$ and $z_p \text{SD}(\hat{\phi})$, respectively. Moreover, $\mathbb{E}[\hat{\phi}] - C = o(\text{SD}(\hat{\phi}))$ because the standard deviation asymptotically dominates the bias. Therefore, the $(1-p)$ - and p -quantiles of $\hat{\phi} - C$ converge to 0 at rate $\Theta(\sqrt{\sum_{j=1}^m N_j^{-1}})$. Since the hypothesis test enforces

$$\frac{\epsilon(N_1, \dots, N_m)}{\sqrt{\sum_{j=1}^m N_j^{-1}}} \rightarrow \infty,$$

$\epsilon(N_1, \dots, N_m)$ converges to 0 more slowly than both tail quantiles. Thus, as every $N_j \rightarrow \infty$, both tail quantiles lie within $[-\epsilon(N_1, \dots, N_m), \epsilon(N_1, \dots, N_m)]$, so $\Pr(|\hat{\phi} - C| > \epsilon(N_1, \dots, N_m)) \leq 1 - [p - (1-p)] = 2(1-p)$. Because p may be chosen arbitrarily close to 1, this probability converges to 0. $\square$

A.3 WHY STRONG SPOOFING IS DIFFICULT: A TOY ARGUMENT

Although 100% long-run detection recall is not formally guaranteed by this method, the following illustrative toy argument suggests a malicious provider may find it difficult to strongly spoof the tamper-detection test. We consider the Temperature Test from Theorem 11 for this example. To avoid making the tampering obvious, suppose that the malicious provider uses the same vocabulary as M and keeps the queries within each experiment IID. Suppose furthermore that, during experiment E_i , the probability that the first generated token returned is t is

$$p_i(t) = \frac{\exp((\tilde{z}_i(t) + b(t))/T)}{\sum_{t' \in S_i} \exp((\tilde{z}_i(t') + b(t'))/T)},$$

where S_i is the set of tokens retained after truncation.

Suppose that the provider strongly spoofs the Temperature Test: in the long-run limit, it passes every valid instance of the test. Choose arbitrary ρ -regular experiments E_i, E_j, E_k, E_ℓ with $A_i = A_j = (a)$, $B_i = B_j = (b)$, $A_k = A_\ell = (c)$, and $B_k = B_\ell = (d)$. By Theorem 11, passing this instance requires

$$\frac{(z_i(a) - z_i(b)) - (z_j(a) - z_j(b))}{\Gamma_i - \Gamma_j} = \frac{(z_k(c) - z_k(d)) - (z_\ell(c) - z_\ell(d))}{\Gamma_k - \Gamma_\ell}.$$

For experiment E_i , the probability ratio is

$$\begin{aligned} \frac{p_i(a)}{p_i(b)} &= \frac{\exp((\tilde{z}_i(a) + b(a))/T)}{\sum_{t' \in S_i} \exp((\tilde{z}_i(t') + b(t'))/T)} \frac{\sum_{t' \in S_i} \exp((\tilde{z}_i(t') + b(t'))/T)}{\exp((\tilde{z}_i(b) + b(b))/T)} \\ &= \exp\left(\frac{\tilde{z}_i(a) - \tilde{z}_i(b) + b(a) - b(b)}{T}\right). \end{aligned}$$

Because $A_i = (a)$ and $B_i = (b)$, it follows that

$$\Gamma_i = \ln\left(\frac{p_i(a)}{p_i(b)}\right) = \frac{\tilde{z}_i(a) - \tilde{z}_i(b) + b(a) - b(b)}{T}.$$

Applying the same calculation to the remaining experiments and subtracting within each experiment pair gives

$$\begin{aligned} \Gamma_i - \Gamma_j &= \frac{(\tilde{z}_i(a) - \tilde{z}_i(b)) - (\tilde{z}_j(a) - \tilde{z}_j(b))}{T}, \\ \Gamma_k - \Gamma_\ell &= \frac{(\tilde{z}_k(c) - \tilde{z}_k(d)) - (\tilde{z}_\ell(c) - \tilde{z}_\ell(d))}{T}, \end{aligned}$$

where $b(a) - b(b)$ cancels from the first subtraction and $b(c) - b(d)$ cancels from the second. Substituting these expressions into the temperature-test equality gives

$$T \frac{(z_i(a) - z_i(b)) - (z_j(a) - z_j(b))}{(\tilde{z}_i(a) - \tilde{z}_i(b)) - (\tilde{z}_j(a) - \tilde{z}_j(b))} = T \frac{(z_k(c) - z_k(d)) - (z_\ell(c) - z_\ell(d))}{(\tilde{z}_k(c) - \tilde{z}_k(d)) - (\tilde{z}_\ell(c) - \tilde{z}_\ell(d))}.$$

Canceling T and taking reciprocals gives

$$\frac{(\tilde{z}_i(a) - \tilde{z}_i(b)) - (\tilde{z}_j(a) - \tilde{z}_j(b))}{(z_i(a) - z_i(b)) - (z_j(a) - z_j(b))} = \frac{(\tilde{z}_k(c) - \tilde{z}_k(d)) - (\tilde{z}_\ell(c) - \tilde{z}_\ell(d))}{(z_k(c) - z_k(d)) - (z_\ell(c) - z_\ell(d))}.$$

Strong spoofing requires the displayed equality to hold for every two valid choices of experiment pairs and token pairs. Every such ratio must therefore equal every other such ratio, so they share one common value λ . Therefore, for every covered choice of i, j, a, b ,

$$(\tilde{z}_i(a) - \tilde{z}_i(b)) - (\tilde{z}_j(a) - \tilde{z}_j(b)) = \lambda((z_i(a) - z_i(b)) - (z_j(a) - z_j(b))).$$

Grouping the terms by experiment gives

$$[\tilde{z}_i(a) - \lambda z_i(a)] - [\tilde{z}_i(b) - \lambda z_i(b)] = [\tilde{z}_j(a) - \lambda z_j(a)] - [\tilde{z}_j(b) - \lambda z_j(b)].$$

Fix a reference experiment E_{i_0} and token t_0 . Taking $a = t$, $b = t_0$, and $j = i_0$ gives

$$\tilde{z}_i(t) - \lambda z_i(t) = [\tilde{z}_i(t_0) - \lambda z_i(t_0)] + [\tilde{z}_{i_0}(t) - \lambda z_{i_0}(t)] - [\tilde{z}_{i_0}(t_0) - \lambda z_{i_0}(t_0)].$$

The first bracket on the right depends only on i , while the difference of the remaining two brackets depends only on t . Defining

$$f(i) := \tilde{z}_i(t_0) - \lambda z_i(t_0), \quad g(t) := [\tilde{z}_{i_0}(t) - \lambda z_{i_0}(t)] - [\tilde{z}_{i_0}(t_0) - \lambda z_{i_0}(t_0)]$$

therefore yields

$$\tilde{z}_i(t) = \lambda z_i(t) + f(i) + g(t).$$

The functions f and g are otherwise generic: $f(i)$ depends only on the experiment index, whereas $g(t)$ depends only on the token.

For $\lambda > 0$, define $c(t) = (g(t) + b(t))/\lambda$. Then

$$\begin{aligned} p_i(t) &= \frac{\exp((\lambda z_i(t) + f(i) + g(t) + b(t))/T)}{\sum_{t' \in S_i} \exp((\lambda z_i(t') + f(i) + g(t') + b(t'))/T)} \\ &= \frac{\exp((\lambda z_i(t) + g(t) + b(t))/T)}{\sum_{t' \in S_i} \exp((\lambda z_i(t') + g(t') + b(t'))/T)} \\ &= \frac{\exp((z_i(t) + c(t))/(T/\lambda))}{\sum_{t' \in S_i} \exp((z_i(t') + c(t'))/(T/\lambda))}. \end{aligned}$$

The served distribution is therefore obtained by applying an effective temperature T/λ , a token bias $c(t)$, and some unspecified truncation rule to the expected model's logits. Under appropriate coverage assumptions on the tested tokens and regularity assumptions on the type of truncation used by the provider, spoofing every valid Temperature Test requires reproducing the expected model configuration's exact behavior for the first generated token returned up to standard decoding parameters, when $\lambda > 0$. As the old adage says, "if it walks like a duck and quacks like a duck, it is a duck." Going by this wisdom, regardless of the internals of the served model configuration, if it externally behaves identically to the expected model configuration up to standard decoding parameters, it would functionally not constitute tampering in the first place, and the malicious provider would have failed in its fundamental objective.